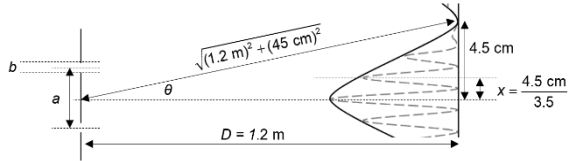


18



$$\sin \theta = \frac{\lambda}{b} \text{ and } \lambda = \frac{ax}{D}$$

$$\sin \theta = \frac{ax}{bD}$$

$$\frac{b}{a} = \frac{x}{D \sin \theta} = \frac{4.5 \times 10^{-2} \div 3.5}{(1.2) \frac{4.5 \times 10^{-2}}{\sqrt{(4.5 \times 10^{-2})^2 + 1.2^2}}} = 0.29$$

19 The electric field between two charged parallel plates is uniform, hence it has the same constant